

I. PREPROCESSING & ALIGNMENT

YuNet 5-Point Alignment

Real-Time ONNX Face Detection
LMEDS Similarity Transform ($s, \theta, \mathbf{t}$)
Expansion: $s_f = 1.50 \times$ (Ear-to-Ear)

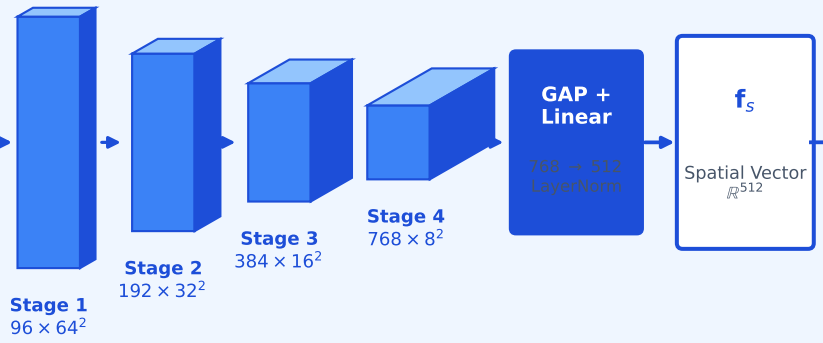
Hann Edge Tapering

Outer Boundary Windowing
 $H_{\text{taper}} = 12$ pixels Transition Band
Suppresses Cartesian Phase Leakage

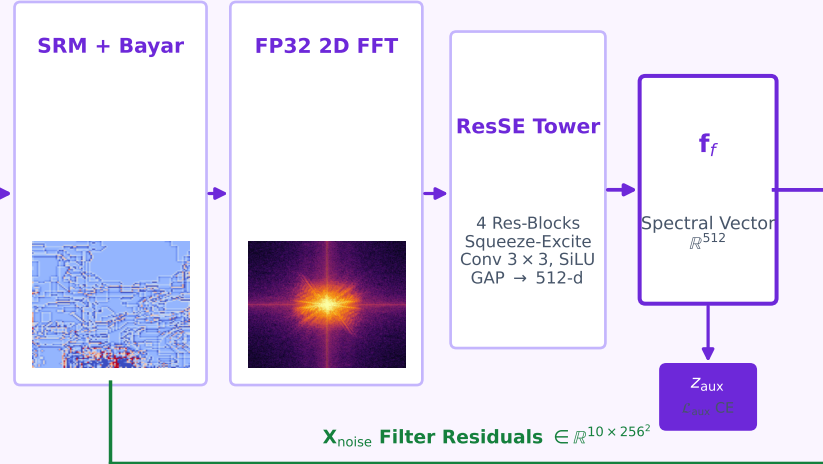


Input Face Crop I
 $\mathbb{R}^{3 \times 256 \times 256}$

II-A. SPATIAL STREAM: ConvNeXt-Small (50.2M)



II-B. SPECTRAL STREAM: ResSE Tower (2.99M)



III. SNR-ADAPTIVE GATED FUSION

Complementary Feature Fusion

$$\mathbf{f}_{\text{fused}} = [(1 - \mathbf{g}_{\text{eff}}) \odot \mathbf{f}_s \parallel \mathbf{g}_{\text{eff}} \odot \mathbf{f}_f] \in \mathbb{R}^{1024}$$
$$\mathbf{e}_t = (1 - \mathbf{g}_{\text{eff}}) \odot \mathbf{f}_s + \mathbf{g}_{\text{eff}} \odot \mathbf{f}_f \in \mathbb{R}^{512}$$

Gate MLP

Concat $\mathbb{R}^{1024}$
1024 $\rightarrow$ 256
 $\sigma(\cdot) \rightarrow \mathbf{g}$

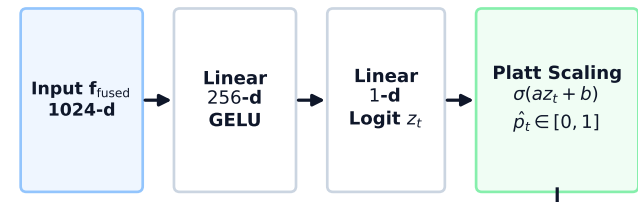
$$\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$$

Noise Power Monitor Circuit

$$P_{\text{noise}} = \frac{1}{|\Omega|} \sum |\mathbf{X}_{\text{noise}}|^2 \rightarrow \gamma = \text{clip}\left(\frac{P_{\text{noise}} - 0.005}{0.025 - 0.005}, 0, 1\right)$$

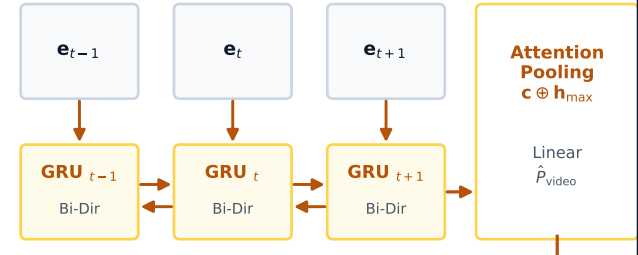
IV. FORENSIC DECISION HEADS & TRIAGE

[BRANCH A] Single-Frame Classifier



Platt Calibration: $a = 0.2783, b = 0.4089$ | Optimal Threshold: $\tau^* = 0.26$
Frame Performance: AUC = 0.8656 | PR AUC = 0.9374 | Precision = 90.51%

[BRANCH B] Spatiotemporal Video Bi-GRU Head



Sequence Performance: AUC = 0.8994 | EER = 18.54% | 2-Layer Bi-GRU ($d_h = 256$)

Input: $[\mathbf{e}_t \parallel \Delta \mathbf{e}_t] \in \mathbb{R}^{1024}$ | Stride $\Delta k = 5$

Zone 1: Clearance
 $\hat{p} < 0.40$ (46.3%)
Auto-Dismissal

Zone 2: Review
[0.40, 0.60] (5.1%)
Expert Triage

Zone 3: Confirmed
 $\hat{p} \geq 0.60$ (48.6%)
Auto-Interdict